ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)
JEE(Main)
FULL SYLLABUS
05-05-2021
JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE [SCORE-II]

PART-1 : PHYSICS — SOLUTION

SECTION-I

1. Ans. (A)

Sol. $\Delta S = nC\ln\frac{T_b}{T} + \frac{mL}{T_b} = 697\ J/K$

2. Ans. (C)

Sol. $m\frac{dx}{dt^2} + \gamma m\frac{dx}{dt} + kx = F_0\cos\omega t$

Now solve

3. Ans. (D)

Sol. $f_r = m\omega^2 r$

$f_s = m\alpha r$

$\tan\theta = \frac{f_s}{f_r}$

4. Ans. (C)

Sol. Check all options : $E = \frac{1}{2}fnRT$

5. Ans. (B)

Sol. $B = \frac{\mu_0 ni}{2}(\underset{0}{\downarrow}\cos\theta_1 + \underset{\frac{\ell}{\sqrt{\ell^2+R^2}}}{\downarrow}\cos\theta_2)$

$\& \ n = \frac{1}{\ell}$

6. Ans. (B)

Sol. Force between two charges will not change due to presence of 3rd charge.

7. Ans. (A)

Sol. $U = -\vec{\mu}_2.\vec{B}_1 \ \& \ B_1 = \frac{\mu_0}{4\pi}\frac{2\mu_1}{r^3}$

$F = -\frac{dU}{dr}$

8. Ans. (D)

Sol. $KE = \frac{1}{2}m\omega^2\left(A^2 - x^2\right)$

$PE = \frac{1}{2}m\omega^2 x^2$

$\therefore \ KE = \frac{1}{2}m\omega^2 A^2 + PE$

9. Ans. (C)

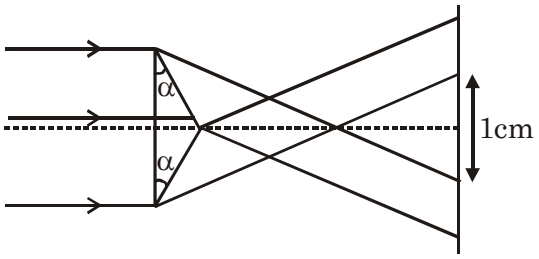
Sol. step 1 : Find equilibrium position x_0.

step 2 : Find U at x_0

step 3 : work done = $U_f - U_i = 0 - U$

10. Ans. (A)

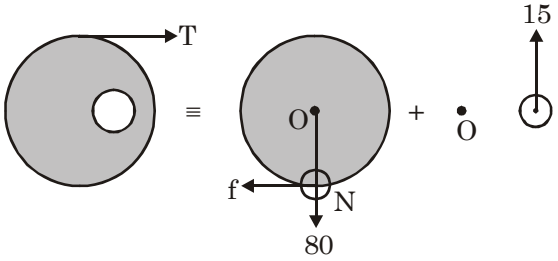
Sol.



Now solve

11. Ans. (A)

Sol.



$15 \times \frac{2a}{5} = T\,2a \qquad \therefore \ T = 3$

12. Ans. (A)

13. Ans. (B)

Sol. $mg - T = ma$

$T = ma \qquad \therefore \ T = mg/2$

$a = g/2$

14. Ans. (C)

Sol. The projectile is initially at semi-minor axis

$\therefore$ Satellite is at equi distance from both focuses.

15. Ans. (B)

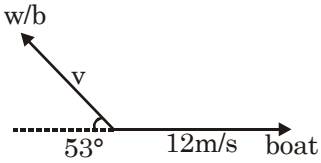
Sol. Phase diff between top & middle is π.

& middle & bottom is π.

$\therefore$ between top & bottom is 2π.

16. Ans. (A)

Sol.



$v \cos 53° = 12 \quad \Rightarrow v \times \frac{3}{5} = 12$

$v = 20$

$\therefore v \sin 53° = 20 \times \frac{4}{5} = 16$ m/s

17. Ans. (D)

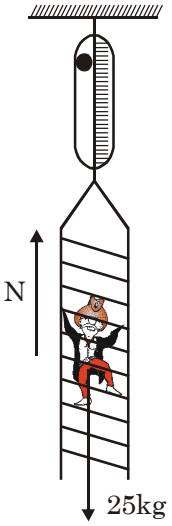
Sol. If the helicopter is at terminal velocity, then it is falling at constant speed. As such, $h = vT$, where v is the terminal velocity. In that case, $\alpha = 1$.

18. Ans. (C)

Sol. Charge on inner most shell is $-\left(\frac{r_{inner}}{r_{outer}}\right)Q$.

19. Ans. (B)

Sol.



$N - 25g = 25 \times 1$

$N = 275$

$\therefore W = 275 + 50 = 325$ N

20. Ans. (A)

Sol. Since the rod will rotate

$\therefore$ velocity will change direction atleast

SECTION-II

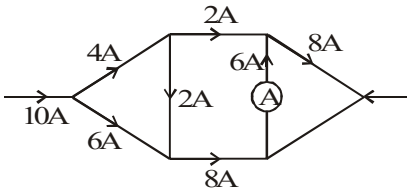
1. Ans. 0.75

Sol. $E = \frac{\sigma}{4\varepsilon_0}$ due to spherical shell.

$E = \int_R^{2R} \frac{\rho 2\pi r^2 dr}{4 \in_0}$, where $\rho = \frac{\rho_0 r}{R}$

2. Ans. 6.00

Sol.



3. Ans. 0.05

Sol. $M_\theta = \frac{1.5°}{0.25°} = \frac{f_0}{f_e}$ & $f_0 + f_e = 35$

4. Ans. 4.00

Sol. $T = 2\pi\sqrt{\frac{I}{\mu B}}$

$\therefore T \propto \sqrt{I}$

5. Ans. 0.50

Sol. $e = \frac{v_f}{v_i} = \frac{\sqrt{2gh}}{\sqrt{4gh}} = \frac{1}{\sqrt{2}}$

6. Ans. 0.18

Sol. $\beta = \frac{\lambda D}{d} = \frac{6563 \times 10^{-10} \times 1m}{3.6 \times 10^{-3}}$

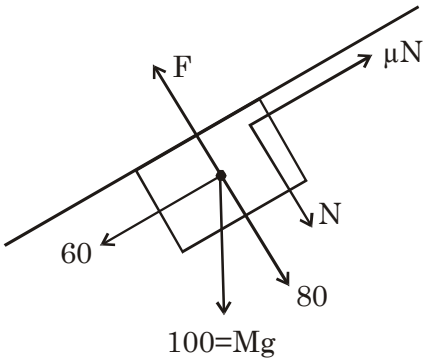
$\beta = 0.0182$ cm

7. Ans. 697.00

Sol. $\Delta S = nC\ell n\frac{T_b}{T} + \frac{mL}{T_b} = 697 J/K$

8. Ans. 200.00

Sol.



$\mu N = 60 \Rightarrow N = 120$

$\Rightarrow F = 200$

9. Ans. 4.47

Sol. $E = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$

$= -\left(2x\hat{i} + 2y\hat{j} + 2z\hat{k}\right)$

$$\vec{E} = -\left(2\hat{i} + 0\hat{j} + 4\hat{k}\right)$$

$$E = \sqrt{20} = 2\sqrt{5} = 4.47$$

10. Ans. 1316.00

Sol. $N + mg = \frac{mv^2}{R}$

$$N + 700 = \frac{70 \times (120)^2}{500}$$

$$N = 1316$$

PART-2 : CHEMISTRY — SOLUTION

SECTION-I

1. Ans. (D)

Sol. No. of spherical nodes = $n - \ell - 1$

$\Rightarrow 3 - 1 - 1 = 1$ for $3p_x$

$\Rightarrow 4 - 2 - 1 = 1$ for $^{4}d_{x^2-y^2}$

2. Ans. (B)

Sol. At this high P = 40 atm

$V_{gas} >> a$

$\therefore \frac{a}{V_m^2} << P$

$\therefore$ equation becomes $P(v_m - b) = RT$

$\therefore T = \frac{P(v_m - b)}{R} = \frac{40(4 - 0.2)}{0.082}$

= 1854 K

3. Ans. (B)

Sol. $K_p = K_c (RT)^{\Delta_g n}$

$K_p > K_c$ if $\Delta_g n > 0$ & $T > \frac{1}{R}$; $\left\{\frac{1}{R} = 12.19\right\}$

$K_p > K_c$ if $\Delta_g n < 0$ & $T < \frac{1}{R}$

4. Ans. (D)

Sol. $Zn(OH)_2 \rightleftharpoons \underset{S}{Zn^{+2}} + \underset{2S}{2OH^-}$

$S = \left(k_{sp}/4\right)^{1/3} = 10^{-4}$

In presence of 0.01 M Zn^{+2}.

$Zn(OH)_2 \rightleftharpoons \underset{S'+0.01}{Zn^{+2}} + \underset{2S'}{2OH^-}$

$k_{sp} = 0.01 \times (2S)^2$

$S = \sqrt{4 \times 10^{-12}/10^{-2} \times 4} = 10^{-5}$

5. Ans. (A)

Sol : Fact or process dependent.

6. Ans. (B)

Sol. $CMC \ \alpha \frac{1}{\text{Vander waal force between hydrophobic part of surfactant}}$

7. Ans. (B)

Sol. cell Rk^n_-

$H_2 \rightarrow 2H^+ + 2e^- \} \times 5$

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O \} \times 2$

$5H_2 + 2MnO_4^- + 16H^+ \rightarrow \underset{\text{anode}}{10H^+} + 2Mn^{+2} + 8H_2O$

$E_{cell} = (1.51 - 0) - \frac{0.06}{10} \log \frac{[Mn^{+2}]^2 [H^+]^{10}_{anode}}{[MnO_4^-]^2 . [H^+]^{16}_{cathode} \left(P_{H_2}\right)^5}$

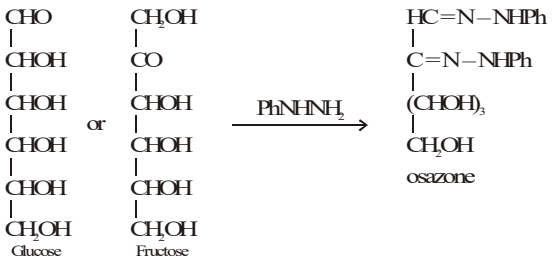
$= 1.51 - \frac{0.06}{10} \log \frac{(10^{-1})^2 (10^{-1})^{10}}{(0.1)^2 (1)^{16} \times 1}$

$= 1.51 - \frac{0.06}{10} \log \frac{10^{-12}}{10^{-2}} = 1.51 + \frac{0.06}{10} \times 10$

= 1.57 V

8. Ans. (B)

Sol. D(+)–glucose and D(–)–fructose forms same osazone on reaction with $PhNHNH_2$



9. Ans. (B)

Sol. $CH_3COOCH_2CH_3$

$\xrightarrow[(ii)H_3O^+]{(i)LiAlH_4/Ether} CH_3CH_2OH + CH_3CH_2OH$

10. Ans. (C)

Sol. $CH_3CH_2OH \xrightarrow[NaOH]{I_2} \underset{\text{Yellow ppt}}{CHI_3 \downarrow} + HCOONa$

$CH_3OH \xrightarrow[NaOH]{I_2}$ No ppt

11. Ans. (D)

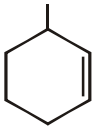
Sol. (A) $CH_3CH_2COOH \xrightarrow[ii)H_3O^+]{i)LiAlH_4/ether} CH_3CH_2CH_2OH$

(B) $CH_3COCl \xrightarrow{R_2CuLi} CH_3COR$

(C) CH_3COOH (vapours) $\xrightarrow[\Delta]{MnO} CH_3COCH_3$

(D) $CH_3CN \xrightarrow[H_3O^+]{SnCl_2/HCl} CH_3CHO$

12. Ans. (C)

Sol. Compound  has non super imposable mirror image.

13. Ans. (A)

Sol. NBS carryout allylic substitution of H atom by free radical mechanism.

14. Ans. (B)

Sol. CN^- is strong field ligand due to which Fe^{+3} configuration will be t_{2g}^5 hence 1 unpaired electron whereas in $[FeF_6]^{3-}$, F^- is weak field ligand hence due to weak field splitting, configuration will $t_{2g}^3 e_g^2$ i.e. 5 unpaired electrons.

15. Ans. (A)

Sol. The intermediate formed in reaction of tetrahalides with H_2O is sp^3d hybridised and carbon does NOT have vacant d-orbitals.

16. Ans. (D)

Sol. H_2O : triatomic and hence planar

BrF_3 :Bent T-shaped

XeF_4: Square planar

17. Ans.(C)

Sol. $Li_2O + H_2O \rightarrow LiOH$,

$KO_2 + H_2O \rightarrow KOH + H_2O_2 + O_2$

$Na_2O_2 + H_2O \rightarrow NaOH + H_2O_2$

18. Ans.(C)

Sol. ZnS - White

PbS - Black

HgO - Yellow

19. Ans. (A)

Sol. Cinnabar-HgS, Calamine-$ZnCO_3$

Carnallite-$KCl.MgCl_2.6H_2O$

Zincite-ZnO

20. Ans.(B)

Sol. $Ca_3P_2 + H_2O \rightarrow Ca(OH)_2 + PH_3$

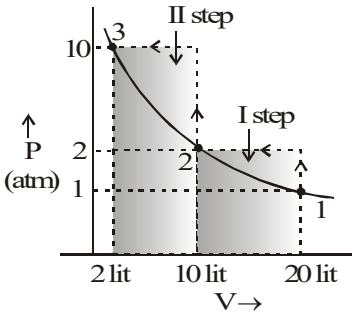
$Mg_3N_2 + H_2O \rightarrow Mg(OH)_2 + NH_3$

$CaC_2 + H_2O \rightarrow Ca(OH)_2 + C_2H_2$

SECTION-II

1. Ans. 100.00

Sol.



If vol. is 10 lit

$$P_2 = \frac{1\times 20}{10} = 2\,atm$$

If V = 2 lit

then $P_3 = \frac{2\times 10}{2} = 10\,atm$

W = (20–10) × 2 + (10–2) × 10

= 100 lit - atm

2. Ans. 2.65

Sol. $\alpha = \sqrt{\frac{k_a}{c}} = \sqrt{\frac{6.4\times 10^{-5}}{10^{-2}}}$

$= \sqrt{6.4\times 10^{-3}} = 8\times 10^{-2} = 0.08$

$\pi = MRT(1+0.08) = 0.01\times 0.082\times 300\times 1.08$

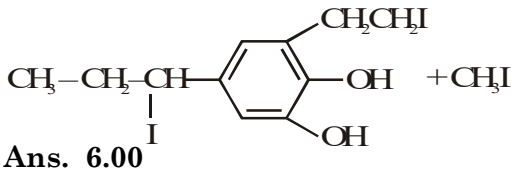
= 0.265 atm

Hence x = 0.265

x × 10 = 2.65

3. Ans. 3.00

Sol.



4. Ans. 6.00

5. Ans. 1.00

Sol. Iron has +1 oxidation state in brown ring complex $[Fe(H_2O)_5(NO)]SO_4$

6. Ans. (324.00)

7. Ans. (–115.80)

8. Ans. (2.25)

9. Ans. (4.00)

10. Ans. (4.00)

PART-3 : MATHEMATICS — SOLUTION

SECTION-I

1. Ans. (B)

Sol. $y^2 = 4(x+1)$ $\qquad x^2 = 4(y+1)$

$y = m(x+1) + \dfrac{1}{m}$ $\qquad y + 1 = mx - m^2$

$\Rightarrow m = 1$

$\Rightarrow$ equation of common tangent is

$x + y + 2 = 0$

Area = 2

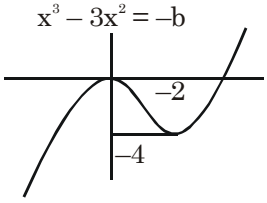
2. Ans. (D)

Sol. $f(x) = x^3 + ax^2 + b$

$f'(x) = 3x^2 + 2ax)_{x=2} = 0$

$12 + 4a = 0$

$a = -3$



$-4 < -b < 0$

$\Rightarrow 0 < b < 4$

3. Ans. (C)

Sol. Equation of normal at any point

$P\left(ct, \dfrac{c}{t}\right)$ on $xy = c^2$ is $xt^3 - yt - ct^4 + c = 0$

If it passes through Q(h, k) then

$ct^4 - ht^3 + kt - c = 0$

$$\Sigma x_1 = \Sigma ct_1 = h, \Sigma y_1 = \sum \frac{c}{t} = k, x_1x_2x_3x_4 = -c^4$$

$$\frac{\Sigma x_1 \Sigma y_1}{x_1x_2x_3x_4} = -\frac{hk}{c^4}$$

If h = 3, k = 2, c = 2

$$-\frac{hk}{c^4} = \frac{-6}{16} = -\frac{3}{8}$$

4. Ans. (A)

Sol. Tangent

$$\frac{x\cos\theta}{12} + \frac{y\sin\theta}{9} = 1$$

A(12secθ, 0) B(0,9cosecθ)

$$m = AB = \sqrt{144\sec^2\theta + 81\operatorname{cosec}^2\theta}$$

$$= \sqrt{225 + \left(12\tan\theta - 9\cot\theta\right)^2 + 216}$$

= 21

Normal

$$\frac{12x}{\cos\theta} - \frac{9y}{\sin\theta} = 63$$

$$P(0,0) = M = \frac{63}{\sqrt{144\sec^2\theta + 81\operatorname{cosec}^2\theta}}$$

Mm = 63

5. Ans. (B)

Sol. $\beta + \gamma = \dfrac{\pi}{2}$

$\cos\beta = \sin\gamma$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\cos^2\alpha = 0$

$(\cos\alpha + \sin\beta + \sin\gamma)^2$

$= \sin^2\beta + \sin^2\gamma + 2\sin\beta\sin\gamma$

$= 1 + \sin 2\beta$

= 2 max value.

6. Ans. (D)

Sol. expand determinant

7. Ans. (A)

Sol. x = 1 $\qquad$ y = –1 $\qquad$ z = 1

8. Ans. (D)

Sol. $x^2 + \dfrac{1}{x^2} = 2\cos\theta$

$\cos\theta = 1$ $\qquad x^2 = 1$

$\theta = 0, 2\pi$ $\qquad x = \pm 1$

9. Ans. (D)

Sol. $\dfrac{\Sigma x_i^2}{n} \geq \left(\dfrac{\Sigma x_i}{n}\right)^2 \Rightarrow \dfrac{200}{n} \geq \dfrac{(40)^2}{n^2} \Rightarrow n > 8$

10. Ans. (A)

Sol. Mode = 3 median – 2 mean

= 3 × 26 – 2 × 25 = 78 – 50 = 28

11. Ans. (B)

Sol.

p	q	$p \leftrightarrow q$	$\sim p$	$\sim p \leftrightarrow q$
T	T	T	F	F
T	F	F	F	T
F	T	F	T	T
F	F	T	T	F

12. Ans. (A)

Sol. by fundamental principal of counting 2^{n^2-n}.

13. Ans. (A)

Sol. $U_{10} = \int_0^{\pi/2} x^{10}\sin x\,dx$

$= \left[-x^{10}\cos x\right]_0^{\pi/2} + \int 10x^9\cos x\,dx$

$U_{10} = 10\left(x^9 \sin x\right)_0^{\pi/2} - 90\int_0^{\pi/2} x^8 \sin dx$

$U_{10} = 10\left(\frac{\pi}{2}\right)^9 - 90\, U_8$

$U_{10} + 90U_8 = 10\left(\frac{\pi}{2}\right)^9$

14. Ans. (A)

Sol. $\int_0^1 f(x)dx + \int_0^1 f(1+x)dx + ... + \int_0^1 f(7+x)dx$

$\int_0^1 f(x)dx + \int_1^2 f(x)dx + ... + \int_7^8 f(x)dx$

$= \int_0^8 f(x)dx = 3$

15. Ans. (B)

Sol. $\int_0^1 [f(x) - f(2x)]dx = 5$ ...(i)

$\int_0^2 [f(x) - f(4x)]dx = 10$ ...(ii)

replace x by 2t in (ii)

x = 2t

dx = 2dt

$2\int_0^1 [f(2t) - f(8t)]dt = 10$ ...(iii)

2 × equation (i) + equation (iii)

$2\int_0^1 [f(x) - f(8x)]dx = 20$

16. Ans. (B)

Sol. $\Rightarrow \int_0^a f(a-x)g(a-x)h(a-x)dx$

$= k\int_0^a f(x)g(x)dx$

$\Rightarrow \int_0^a f(x)g(x)\cdot\left(\frac{3h(x)-5}{4}\right)dx = k\int_0^a f(x)g(x)dx$

$\Rightarrow \frac{3}{4}\int_0^a f(x)g(x)h(x)dx - \frac{5}{4}\int_0^a f(x)g(x)dx$

$= k\int_0^a f(x)g(x)dx$

$\Rightarrow \frac{3}{4}\left[k\int_0^a f(x)g(x)dx\right] - \frac{5}{4}\int_0^a f(x)g(x)dx$

$= k\int_0^a f(x)g(x)dx$

$\Rightarrow \left(\frac{3}{4}k - \frac{5}{4} - k\right)\int_0^a f(x)g(x)dx = 0$

$\frac{3}{4}k - \frac{5}{4} - k = 0$

k = – 5

17. Ans. (C)

Sol. $\alpha = \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3}$

$\beta = \frac{1}{n^4}\left[\left(\frac{n(n+1)}{2}\right)^2 - \frac{n(n+1)(2n+1)}{6}\right]$

$= \frac{n^2(n+1)^2}{4n^4} - \frac{n(n+1)(2n+1)}{6n^4}$

$\beta = \frac{1}{4}$

$\Rightarrow 4\beta = 3\alpha$

18. Ans. (D)

Sol. $\lim_{x\to 0} f(x) = a^3$ independent of x

19. Ans. (D)

Sol. $\lim_{x\to\frac{1}{\sqrt{2}}} f(x) - 2$ must be zero for finite limit.

$\therefore \lim_{x\to\frac{1}{\sqrt{2}}} f(x) - 2 = 0$

$\lim_{x\to\frac{1}{\sqrt{2}}} f(x) = 2$

$\therefore \lim_{x\to\frac{1}{\sqrt{2}}} e^{f(x)} = e^2$

20. Ans. (A)

Sol. $\left(\frac{3x}{5}\right)^2 + \left(\frac{4y}{5}\right)^2 = 1$

$9x^2 + 16y^2 = 25$

Point of intersection with $y = x^2$ is (1,1)

$\left(\frac{dy}{dx}\right) = 2x = 2 = m_1 \left|\left(\frac{dy}{dx}\right)_{at\,(1,1)} = \frac{-9}{16} = m_2\right.$

$\tan\theta = \left(\frac{m_1 - m_2}{1 + m_1 m_2}\right) = \left(\frac{2 + \frac{9}{16}}{1 - \frac{9}{16}\cdot 2}\right) = \frac{41}{2}$

$Q = \tan^{-1}\left(\frac{41}{2}\right)$

SECTION-II

1. Ans. 7.00

Sol. $\left.\begin{matrix} a - 4b + c = 0 \\ \& \; ap + bq + c = 0 \end{matrix}\right\} \Rightarrow p = 1, q = -4$

2. Ans. 42.00

Sol. $[\log 1 + \log 4 + \log 9 + ... + \log 100] - 2\log 7!$

$= 2\log(1 \cdot 2 \cdot 3 \cdot ..... \cdot 10) - 2\log 7!$

$= 2\left[\log\frac{10!}{7!}\right] = 2\log 720 = 2\log(10 \times 9 \times 8)$

$= 2[1 + 3\log 2 + 2\log 3]$

$= 2(1 + 3x + 2y) = 2 + 6x + 4y$

$\therefore a = 2, b = 6, c = 4$

$2a + 3b + 5c = 4 + 18 + 20 = 42$

3. Ans. 2.00

Sol. $\vec{s}.(\vec{p}+\vec{q}+\vec{r}) = 16$

$(x+y+z)[\vec{p}\ \vec{q}\ \vec{r}] = 16$

$x + y + z = 2$

4. Ans. 12.00

Sol. Only possibility for max

$$\begin{pmatrix} 2 & & & & & \\ & 2 & & & & \\ & & 2 & & & \\ & & & 2 & & \\ & & & & 2 & \\ & & & & & 2 \end{pmatrix}$$

5. Ans. 48.00

Sol. $f'(3) = 6$

$$\lim_{h\to 0}\frac{f(3+h)-f(3)}{h} = 6$$

$$\lim_{h\to 0}\frac{f(3+h)}{h} = 6$$

$$\Rightarrow \lim_{3x\to 0}\frac{f(3+3x)}{3x}\times 3 + \lim_{5x\to 0}\frac{f(3+5x)}{5x}\times 5$$

$6 \times 3 + 6 \times 5 = 48$

6. Ans. 7.00

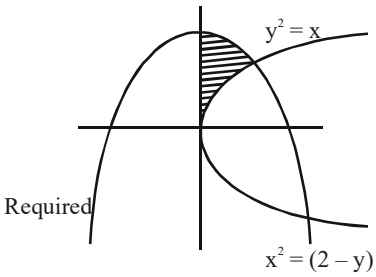
$z^5 = \bar{z}$

$\Rightarrow |z| = 0$ or 1

when $|z| = 0 \Rightarrow z = 0$

$|z| = 1 \Rightarrow z = (1)^{1/6}$

7. Ans. 1.00



Required

$$A = \int_0^1 \left(2 - x^2 - \sqrt{x}\right) dx$$

$$= 2 - \frac{1}{3} - \frac{2}{3} = 2 - 1 = 1$$

8. Ans. –0.50

$$\frac{dy}{dx} + \left(\frac{-x}{1+x}\right)y = \frac{1}{(1+x)}$$

I.F. $= e^{-\int \frac{x}{1+x}dx} = (1+x)e^{-x}$

Solution y(x) is given by

$y \times (1 + x)e^{-x} = \int e^{-x}\,dx = -e^{-x} + C$

$$y(1) = \frac{-1}{2}$$

9. Ans. 165.00

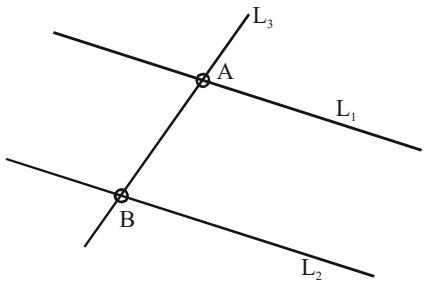
${}^{10}C_8 + {}^9C_7 + {}^8C_6 + ... + {}^4C_2 + {}^3C_1 + {}^2C_0$

$= {}^3C_0 + {}^3C_1 + {}^4C_2 + ... + {}^{10}C_8$

$= {}^4C_1 + {}^4C_2 + ... + {}^{10}C_8$

$= {}^{11}C_8 = \frac{11\times 10\times 9}{3\times 2} = 165$

10. Ans. 78.00



A(5 + 3λ, 7 – λ, –2 + λ)

B(3 + 3μ, 3 + 2μ, 6 + 4μ)

direction of

AB = <2 + 3λ – 3μ, 4 – λ – 2μ, –8 + λ – 4μ>

AB is perpendicular to L_2

$11\lambda - 29\mu - 18 = 0$ ...(1)

Also $\frac{2+3\lambda-3\mu}{2} = \frac{4-\lambda-2\mu}{7} = \frac{-8+\lambda-4\mu}{-5}$

$\Rightarrow \lambda + m + 2 = 0$ ...(2)

from (1) and (2) $\lambda = -1$

$\mu = -1$

$\Rightarrow$ A(2,8,–3) and B(0,1,2)

$(AB)^2 = 4 + 49 + 25 = 78$